Python
Week 4

Functions & Lists

Objective: Organizing your code and storing multiple pieces of data

What You'll Learn:

- What functions are
- How to create and call your own functions
- How to use lists to store multiple values

Do Now (Week 3 Review)

Write a program that:

1. Uses a for loop to count from 1 to 15.
2. If the number is divisible by 3, print "Fizz".
3. Otherwise, print the number.

```
for i in range(1, 16):
    if i % 3 == 0:
        print("Fizz")
    else:
        print(i)
```

I. Introduction to Functions

What is a function?

A function is a block of code that performs a specific task. Instead of writing the same code over and over, you can write it once and use it whenever you need it.

Why use functions?

- Reuse code without repeating yourself
- Keep your programs organized
- Make your code easier to read and understand

You've already used functions! These are built-in functions that Python provides. Today, you'll learn how to create your own!

```python
print("Hello!")
len("Python")
input("What's your name? ")
```

True or False?

A. A function lets you write code once and use it many times.
B. You have to rewrite the same code every time you want to use it.
C. print() is an example of a function in Python.

True or False?

A. A function lets you write code once and use it many times.
   a. Answer: True
B. You have to rewrite the same code every time you want to use it.
   a. Answer: False
C. print() is an example of a function in Python.
   a. Answer: True

II. How to Write Functions

Structure of a Function

```
def greet():
    print("Hello!")
```

def => starts a function
greet => the name of the function
() => holds inputs (parameters)
print() => code that runs when the function is called

To use the function:

```python
greet()
```

Calling a function => tells Python to run the code inside the function

Multiple Choice: What does the following code print out?

A. 5 + 3
B. 8
C. 53
D. Error

```
def add_numbers():
    print(5 + 3)

add_numbers()
```

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    print(5 + 3)

add_numbers()
```

Try it: Write and call a function that prints out your name.

```
def greet():
    print("Hello!")
```

```python
def print_name():
    print("Alex")

print_name()
```

III. Lists

What is a list?

A list stores multiple values in one variable.

Why use lists?

- Store many pieces of data together
- Easily add, remove, or change items
- Use loops to go through every item

Instead of creating separate variables:

```
fruit1 = "Apple"
fruit2 = "Banana"
fruit3 = "Orange"
```

We can use a list:

```
fruits = ["Apple", "Banana", "Orange"]
```

List Structure

```
fruits = ["Apple", "Banana", "Orange"]
```

fruits => name of the list
[ ] => creates a list
"Apple", "Banana", "Orange" => items inside the list

Accessing Items

```python
fruits = ["Apple", "Banana", "Orange"]

print(fruits[0])
```

Output

```
Apple
```

Index => the position of an item in a list

Remember:

- Lists start counting at 0
- First item = index 0
- Second item = index 1
- Third item = index 2

Try it: Create a list of your favorite foods and print the list.

Challenge: Print only the first item in the list.

```python
foods = ["Pizza", "Tacos", "Pasta"]

print(foods)
```

```python
print(foods[0])
```

Now it's your turn!

Write a Program that:

1. Creates a list of 3 favorite movies.
2. Uses a function to print each movie in the list.

Classwork Answer:

```
def print_movies(movie_list):
    for movie in movie_list:
        print(movie)

movies = ["Cars", "Toy Story", "Finding Nemo"]

print_movies(movies)
```

Thanks!

Have questions? Reach out:
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